

## Coding & Solution

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| <b>Date</b>          | 15 July 2026                        |
| <b>Team ID</b>       | SWUID2026217449                     |
| <b>Project Name</b>  | Rising Waters - Flood Prediction ML |
| <b>Maximum Marks</b> | 5 Marks                             |

### Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

#### Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

### Solution Summary

| Field                                | Details                                                                                                                                     |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Repository Link / URL</b>         | <a href="https://github.com/Sunil9969/Rising-Waters-Flood-Prediction-ML">https://github.com/Sunil9969/Rising-Waters-Flood-Prediction-ML</a> |
| <b>Programming Language(s)</b>       | Python                                                                                                                                      |
| <b>Framework(s) Used</b>             | Flask, Scikit-learn, XGBoost                                                                                                                |
| <b>Key Features Implemented</b>      | Flood risk prediction (High/Low), 4 ML model comparison, Flask web interface, real-time prediction form                                     |
| <b>Pending / Incomplete Features</b> | Cloud deployment on IBM Cloud (planned for future phase)                                                                                    |
| <b>Setup / Run Instructions</b>      | 1. Clone repo 2. pip install -r requirements.txt 3. python app.py<br>4. Open <a href="http://127.0.0.1:5000">http://127.0.0.1:5000</a>      |

**Code Quality Checklist**

| S.No | Criteria                                               | Status (Yes / No) |
|------|--------------------------------------------------------|-------------------|
| 1    | Code is modular and organized into functions / classes | Yes               |
| 2    | Meaningful variable and function names are used        | Yes               |
| 3    | Code includes comments / documentation where necessary | Yes               |
| 4    | Error handling is implemented for critical operations  | Yes               |
| 5    | The application runs without critical errors           | Yes               |
| 6    | Code is committed to a version control repository      | Yes               |

**Additional Notes / Comments:**

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